SEVA INNOVATION CHALLENGE 2026 • IDEA LAB, RAMDEOBABA UNIVERSITY, NAGPUR

Multimodal Machine Health Monitoring and Anomaly Detection

A low-cost edge system that watches a machine's heat, vibration and sound, and warns operators and maintenance teams of abnormal behaviour early.



Working prototype: live dashboard replaying recorded thermal, vibration and sound data

Project at a glance

- **Team:** 5 students, B.Tech Artificial Intelligence & Cyber Security
- **Guide:** Prof. Ashwini Lokhande
- **Project code:** 24CS03PR0504 (2026-27)
- **Signals:** thermal video, vibration, sound
- **Runs on:** Raspberry Pi 4 edge computer

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The Problem

What real challenge in society, government service or industry are you solving?

What is the problem?

Motors, pumps, bearings and gearboxes show changes in heat, vibration and sound before they fail. Most plants check these signals separately, by hand or on a schedule, so early warning signs are missed until the machine breaks down.

Who is affected, and where?

Machine operators, maintenance teams and plant owners in manufacturing across India, especially small and medium units that cannot afford commercial condition-monitoring systems.

Evidence

Unplanned downtime costs the world's 500 largest companies about 11% of revenue, US$1.4 trillion a year, up from 8% in 2019.
Source: Siemens, The True Cost of Downtime 2024 (survey of 181 plant professionals).

Current gaps

Commercial systems are costly and usually watch one signal. Manual rounds are periodic, need an expert to read raw spectra, and miss faults that show up in another sensor.

7.3 crore

MSME establishments in India (NSSO ASUSE 2023-24): the plants we design for

Why it matters

A sudden failure stops production, damages equipment and puts workers at risk. Early warning turns it into a planned repair.

3 Our Solution & Innovation

What have you built or designed, how does it work, and what is new about it?

Our solution in brief

Software that takes a machine's thermal video, vibration signal and sound, extracts engineering features, runs a separate trained model for each signal, and shows the results live in a browser. It runs on a Raspberry Pi 4 edge computer.

How it works

1 Input

Thermal video, vibration (20 kHz) and sound (16 kHz)

›

2 Process

Features, 3 trained models, anomaly checks, confidence gate

›

3 Output

Live result, confidence and alerts for each signal

What makes it new?

	Existing approach	Our approach
Cost	Paid systems or expert surveys	**₹5,500 edge computer**
Speed / time	Periodic manual rounds	**Continuous, every second**
Reach / access	Usually one signal	**Heat, vibration and sound**
Ease of use	Expert reads raw spectra	**Plain status in a browser**

Key USP

Each signal has its own trained model, and a confidence gate holds back results the model is unsure of, so operators see "not sure" instead of a false alarm.

4 Feasibility & Implementation

How ready is the solution, and what will it take to implement it on the ground?

Current stage: working software prototype, tested on recorded data

Idea — Proof of Concept — **Working Prototype** — Field Pilot — Ready to Deploy

Technology

Python (FastAPI), PyTorch, scikit-learn, React dashboard. Runs today on a Raspberry Pi 4. 199 automated tests pass. Next: thermal camera, vibration sensor and microphone.

Estimated cost

Edge computer: about ₹5,500 (Raspberry Pi 4, 4 GB).
Thermal module: about ₹5,900–7,350 (MLX90640).
Vibration sensor and microphone: to be costed.
Software: no licence fee.

Timeline

3 months: live sensors on a lab motor.
6 months: own labelled data; re-test the models.
12 months: pilot on 2–3 shop-floor machines.

Partners

Seeking an MSME or PSU workshop to host a pilot, and the university's mechanical lab for test rigs.
No partner is confirmed yet.

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Impact & Scalability

What change will your solution create, and how can it grow beyond the first location?

98%

Sound anomalies caught

41 of 42 held-out faulty clips; macro-F1 0.88

72.5%

Thermal images classified

51 held-out images, 6 classes; top-3 96.1%

₹5,500

Edge computer runs all 3 models

Raspberry Pi 4; no cloud service needed

Social impact

Fewer sudden breakdowns near workers makes shop floors safer. A plain status instead of raw spectra lets small units without a vibration expert use condition monitoring.

Sustainability

Low-cost hardware and open-source software keep running costs to power and upkeep. Can be offered to MSMEs as a low-cost service, or funded through CSR or cluster programmes.

Scale-up plan

The same software runs on every node, so adding a machine means adding a node. Each new machine type needs its own data to set baselines and retrain the models before results are trusted.

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Team/Individual & Support Required

Who is behind the idea, and what help do you need from the Challenge?

VB — **Vaishnav Balpande**
Roll no. B1-01

SN — **Shreyank Nipane**
Roll no. B4-55

TS — **Tanushri Sapate**
Roll no. B3-44

AP — **Aastha Pusdekar**
Roll no. B4-52

KM — **Kartik Madankar**
Roll no. B3-38

AL — **Prof. Ashwini Lokhande**
Faculty guide

Department of Artificial Intelligence & Cyber Security, Ramdeobaba University, Nagpur

Thank you | Idea Lab, Ramdeobaba University, Nagpur

Support we are seeking

- [x] Mentorship & technical guidance
- [x] Pilot with a government department
- [x] Funding / grant support
- [ ] Incubation & workspace
- [ ] Testing, certification or IP support

Mentoring, a pilot site and funds for sensors would take us from recorded data to live monitoring of real shop-floor machines.